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S2**Started on** Monday, 24 August 2026, 5:30 PM**State** Finished**Completed on** Monday, 24 August 2026, 5:38 PM**Time taken** 7 mins 43 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int count = 0;
8
9      int i = n / 2;
10     count++; // outer initialization / base count
11
12     while (i < n) {
13         count++; // outer loop condition (i < n)
14
15         int j = 1;
16         while (j < n) {
17             count++; // middle loop condition (j < n)
18
19             int k = 1;
20             while (k < n) {
21                 count++; // inner loop condition (k < n)
22                 count++; // c++ body statement
23                 k *= 2;
24             }
25
26             j *= 2;
27             count++; // middle loop update (j = 2 * j)
28         }
29
30         i++;
31         count++; // outer loop update (i++)
32     }
33
34     count++; // final outer loop condition (false)
35
36     printf("%d\n", count);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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